



Types of Matrices

Continuation...

7. Symmetric Matrix

This type of matrix is a square matrix whose transpose is equal to the original matrix.

$$A = \begin{bmatrix} 3 & 2 & 6 \\ 2 & 5 & 4 \\ 6 & 4 & 5 \end{bmatrix}$$

8. Skew-Symmetric Matrix

This type of matrix is a square matrix whose transpose is equal to the negative of the original matrix.

$$A = \begin{bmatrix} 3 & 2 & -6 \\ -2 & 5 & 4 \\ 6 & -4 & 5 \end{bmatrix}$$

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Operation of Matrices

1. Scalar Multiplication

A matrix with dimension $m \times n$ can be multiplied to any scalar number c by multiplying all elements by the scalar number c .

$$A = \begin{bmatrix} 6 & 5 & 4 \\ 2 & 1 & 4 \\ 3 & -2 & -2 \end{bmatrix}$$

Determine $-A$:

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$$-A = \begin{bmatrix} -6 & -5 & -4 \\ -2 & -1 & -4 \\ -3 & 2 & 2 \end{bmatrix}$$

Determine $3A$:

$$3A = \begin{bmatrix} 18 & 15 & 12 \\ 6 & 3 & 12 \\ 9 & -6 & -6 \end{bmatrix}$$

2. Addition of Matrices

If A and B are two matrices of the same order/size/dimensions, the sum of A and B , denoted by $A + B$, is the matrix for which each of its elements is the sum of the corresponding elements of A and B .

Example: Find $A + B$

$$A = \begin{bmatrix} 5 & -2 & 6 \\ -1 & 8 & -3 \end{bmatrix}$$
$$B = \begin{bmatrix} 4 & 7 & 1 \\ 7 & 0 & 4 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 5+4 & -2+7 & 6+1 \\ -1+3 & 8+0 & -3+4 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 9 & 5 & 7 \\ 2 & 8 & 1 \end{bmatrix}$$

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The same addition rule applies for adding row or column matrices or vectors. Listed below are some rules in the addition of matrices:

$$\begin{aligned} A + B &= B + A \\ (A + B) + C &= A + (B + C) \\ A + 0 &= A \\ A + (-A) &= 0 \end{aligned}$$

$$\begin{aligned} c(A + B) &= cA + cB \\ (c + k)A &= cA + kA \\ c(kA) &= (ck)A \\ 1A &= A \end{aligned}$$

3. Subtraction of Matrices

If A and B are matrices having the same order, then the difference of A and B , denoted by $A - B$, is defined as $A - B = A + (-B)$.

Example: Find $A - B$.

$$A = \begin{bmatrix} 4 & -3 \\ -5 & -1 \\ 3 & 8 \end{bmatrix}$$
$$B = \begin{bmatrix} 4 & 3 \\ 7 & -5 \\ 8 & 2 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} 4 & -3 \\ -5 & -1 \\ 3 & 8 \end{bmatrix} + \begin{bmatrix} -4 & -3 \\ -7 & 5 \\ -8 & -2 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 0 & -6 \\ -12 & 4 \\ -5 & 6 \end{bmatrix}$$

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The same subtraction rule applies for getting the difference of two row or column matrices or vectors.

4. Multiplication of Matrices

If you have two matrices, A and B , the product AB is defined when the number of columns of A is equal to the number of rows of B . Supposed that A is a matrix of order $m \times p$ and B is a matrix of order $p \times n$, then the product of A and B , denoted by AB , is the $n \times n$ matrix for which the element in the i th row and the j th column is the sum of the products formed by multiplying each elements in the i th row of A by the corresponding element in the j th column of B .

Listed below are some rules in the multiplication of matrices:

- Matrix multiplication is not commutative
- Matrix multiplication is associative and distributive with respect to addition of matrices
- $AB = 0$ does not necessarily mean that $BA = 0$ or that $A = 0$ or $B = 0$

Example: Find DC . *Excel Review Center*

$$D = \begin{bmatrix} 2 & -3 \\ 4 & -1 \\ 1 & 5 \end{bmatrix}$$
$$C = \begin{bmatrix} -4 & 1 & 3 \\ -1 & 2 & 5 \end{bmatrix}$$

D is a 3×2 matrix and C is a 2×3 matrix. The product DC can be obtained because the number of columns of D (2 columns) is equal to the number of rows of C (2 rows). The product DC will be a 3×3 matrix.

$$DC = \begin{bmatrix} 2(-4) + (-3)(-1) & 2(1) + (-3)(2) & 2(3) + (-3)(5) \\ 4(-4) + (-1)(-1) & 4(1) + (-1)(2) & 4(3) + (-1)(5) \\ 1(-4) + 5(-1) & 1(1) + 5(2) & 1(3) + 5(5) \end{bmatrix}$$

$$DC = \begin{bmatrix} -5 & -4 & -9 \\ -15 & 2 & 7 \\ -9 & 11 & 28 \end{bmatrix}$$

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